EXACT RELATIVE DIMENSION D DESCENDS TO A FINITE LEVEL

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Limit dimension. Limits, Lemma lemma-limit-dimension (same chapter)
- Openness of bounded fibre dimension. Morphisms, Lemma morphisms-lemma-openness-bounded-dimension-fibres-finite-presentation
- Limit contained in constructible. Limits, Lemma lemma-limit-contained-in-constructible (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02G0: sources/stacks-chow/tags.txt, source lines 3211–3211. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f has relative dimension d, and*
(2) *f_0 is locally of finite presentation,*

then there exists an $i \geq 0$ such that f_i has relative dimension d.

Proof. By Lemma 05M5 we may assume all fibres of f_0 have dimension $\leq d$. By Morphisms, Lemma 02G0 the set $U_0 \subset X_0$ of points $x \in X_0$ such that the dimension of the fibre of $X_0 \to Y_0$ at x is $\leq d - 1$ is open and retrocompact in X_0. Hence the complement $E = X_0 \setminus U_0$ is constructible. Moreover the image of $X \to X_0$ is contained in E by Morphisms, Lemma 02FY. Thus for $i \gg 0$ we have that the image of $X_i \to X_0$ is contained in E (Lemma 05F4). Then all fibres of $X_i \to Y_i$ have dimension d by the aforementioned Morphisms, Lemma 02FY. □

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